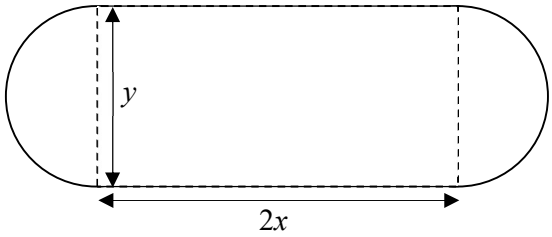
RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 1 (Delta)
Topic(s): Calculus I
Differentiation, Applications, Maclaurin and Graphing (Questions)

1 [It is given that a sphere of radius r has surface area $4\pi r^2$ and volume $\frac{4}{3}\pi r^3$.]

A capsule is to be constructed using the curved surface of a cylinder and the curved surface of a hemisphere at each end. The length of the cylinder is $2x$ metres and the diameter of each hemisphere is y metres. The cross-sectional view of the capsule is shown in the diagram below.



It is given that the construction cost per square metre for the cylinder is \$3k and that for the hemispheres is \$5k, where k is a constant.

Given that the volume of the capsule is π m^3, show that the construction cost of the capsule, C satisfies $C = \frac{12\pi k}{y} + 3\pi k y^2$. [4]

Hence find, using differentiation, the exact values of x and y that give a minimum cost for its construction. [4]

2 The curve C has parametric equations

$$x = t^2 + 2, \quad y = t^3, \text{ where } t \in \mathbb{R}.$$

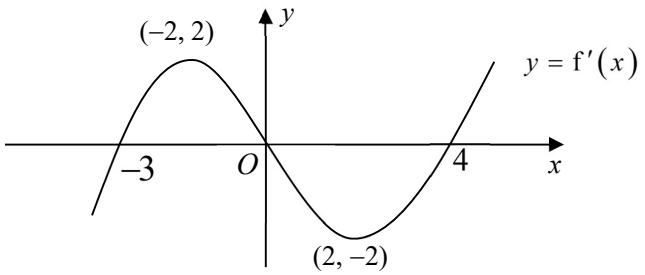
(a) Sketch C. [1]

(b) Find the equation of the tangent to C at the point $(6, 8)$. [2]

(c) The tangent to C at the point $(6, 8)$ meets the curve C again at point P. Find the coordinates of point P. [3]

(d) The normal to C at the point $(m^2 + 2, m^3)$ meets the x- and y-axes at the points Q and R respectively. Given that the point F is the mid-point of QR, find the coordinates of F in terms of m. [4]

3 **(a)** The diagram shows the **derivative** graph of $y = \mathrm{f}(x)$.



Justifying your answers, find the range of values of x for which the graph $y = \mathrm{f}(x)$ is

(i) decreasing, [2]

(ii) increasing and concave downwards. [2]

(b) [It is given that the volume and surface area of a sphere of radius r is $\frac{4}{3}\pi r^3$ and $4\pi r^2$ respectively.]

Air is pumped into a spherical balloon at a constant rate of $12\pi\ \text{cm}^3$ per second.

(i) Find the rate of increase of the balloon's surface area when the volume is $\frac{256}{3}\pi\ \text{cm}^3$. [4]

(ii) Show that the rate of increase of the balloon's radius is inversely proportional to the surface area of the balloon. [1]

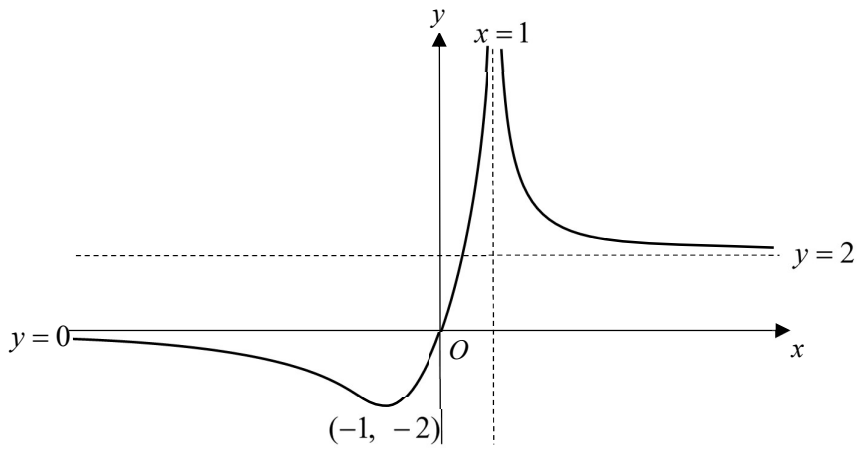
4 The curve C has equation given by $y = \dfrac{1+3x-18x^2}{9x+3}$.

(a) Without the use of a calculator, find the range of values that y can take. [4]

(b) Sketch the graph of C indicating clearly its asymptotes and stationary points. [3]

(c) State a sequence of transformations that will transform the curve C to the curve with equation $y = \dfrac{1+3(2x-1)-18(2x-1)^2}{9(2x-1)+3}$. [2]

5 The graph of $y = \mathrm{f}(x)$ is given below. It has one vertical asymptote at $x = 1$ and two horizontal asymptotes $y = 0$ and $y = 2$. The graph passes through the origin O and has a turning point at $(-1,\ -2)$.



(a) On separate diagrams, sketch the following graphs, indicating the coordinates of the points where the graphs cross the axes, the turning points, and the equations of any asymptotes if possible.

(i) $y = \dfrac{1}{\mathrm{f}(x)}$ [3]

(ii) $y = \mathrm{f}'(x)$ [3]

(iii) $y = \mathrm{f}(|x-1|)$ [2]

(b) State the range of values of a such that $\mathrm{f}(|x-1|) = a$ has only positive root(s). [1]

6 **(a)** State a sequence of transformations which maps the graph of $(x-1)^2 + (y+1)^2 = 1$ to the graph of $\left(\dfrac{x-a}{a}\right)^2 + (2y)^2 = 1$, where a is a positive constant. [3]

(b) The graph of $y = \mathrm{f}(x)$ has axial intercepts at $(a,0)$ and $(0,b)$, and an asymptote at $y = b$, where a and b are positive constants.

State, if it is possible to do so,

(i) the equations of any asymptote(s) and coordinates of axial intercept(s) for the graph of $y = \dfrac{1}{\mathrm{f}(x)}$, [3]

(ii) the coordinates of all axial intercept(s) for the graph of $y = \mathrm{f}(|x|)$. [2]

7 Given that $y = \dfrac{\ln\sqrt{1-x}}{2+x}$, where $-1 \le x < 1$, show that

$$2y + 2(2+x)\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{1-x} = 0 .$$ [2]

(i) By further differentiation, find the Maclaurin's series for y up to and including the term in x^3. [5]

(ii) Verify that the same result can be obtained if the standard series expansions are used. [3]

(iii) By substituting $x = -1$ to your result, show that $\ln 2 \approx \dfrac{a}{b}$, where a and b are constants to be found. [2]

8 Given that θ is a sufficiently small angle, show that

$$\frac{1}{\sin 2\theta + \cos\theta} \approx 1 + a\theta + b\theta^2,$$

where a and b are constants to be determined. [4]